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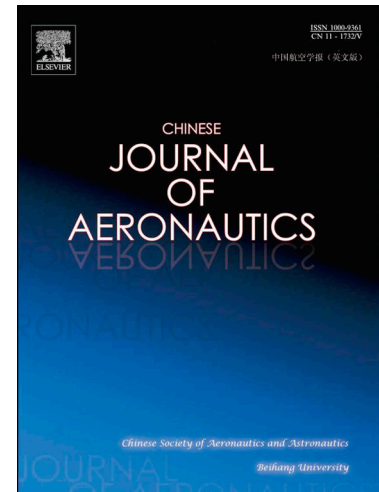
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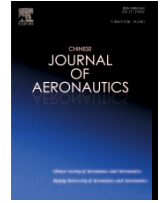
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SHAP-based data mining for high-dimensional design space of high-pressure turbine stage

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Abstract

To clarify the influence of various design variables on turbine stage performance in a high-dimensional design space, SHapley Additive exPlanations (SHAP), a data mining method, is applied to the 93-dimensional design space of the first-stage of high-pressure turbine of the GE-E3 engine. Besides the global performance analysis based on scalar output (0-dimension data), advanced methods are developed to analyze the impact of significant variables on spanwise performance distributions (1-dimension data) and meridional physical field contours (2-dimension data). These methods enable the localization and visualization of specific physical regions and parameters within the turbine stage that are influenced by the significant design variables improving overall performance. Leveraging SHAP's local interpretability, typical samples are selected for attribution analysis to reveal the contribution of individual design variables to performance variations. The results show that significant variables affecting stage efficiency include effective outlet angles, three-dimensional stacking parameters of stator, and spline control points near the suction-side leading edge. Based on the data mining of the GE-E3 dataset, design guidelines are summarized. The final optimized design following these guidelines achieves a 0.65% improvement in stage efficiency.

Keywords: High-pressure turbine stage; High-dimensional design space; Aerodynamic performance; Data mining; SHapley Additive exPlanations (SHAP)

1. Introduction¹

The aerodynamic design of turbines has long been a focus of research among scholars worldwide. With modern aero-engines pursuing higher performance and lower energy consumption, optimizing the aerodynamic profiles of turbine blades to reduce cascade losses and enhance stage efficiency poses significant challenges. Computational Fluid Dynamics (CFD) has substantially reduced the cost of aerodynamic design. Furthermore, the widespread application of optimization algorithms^{1,2} and deep learning techniques^{3,4} in the automatic optimization and flow field prediction of turbomachinery has provided powerful technical support for advanced aerodynamic design.

In recent years, with the increasing number of optimization objectives and design variables, the amount of sample data has grown significantly⁵. Various data mining techniques have been applied to the aerodynamic design of turbine blades. On one hand, data mining enables full utilization of the large volume of data generated during the optimization process, helping to reveal the relationships between design variables and performance metrics, and to identify common features within the optimal solution set. On the other hand, data mining can be employed to extract design guidelines from the design space or the Pareto front, thereby reducing the heavy reliance on designer experience in turbine development and mitigating uncertainties in the design process.

Li and Zheng⁶ reviewed various data mining methods used in aerodynamic optimization design of turbomachinery, including Analysis of Variance (ANOVA), Self-Organizing Maps (SOM), and Proper Orthogonal Decomposition (POD). Through these data mining methods, significant variables can be identified first. For example, ANOVA obtains sensitivity by calculating the contribution of each variable and its coupling effects to the total variance of the objective, and it also characterizes how the variables influence overall performance metrics. Baert et al.⁷ applied ANOVA and other data mining methods to extract knowledge from the design space of a turboprop engine blade, which guided the setting of design variables and efficiently completed a multivariable blade optimization involving 111 design variables. Chiba et al.⁸ conducted ANOVA on a wing dataset and found that the coupling effects among variables were an important factor influencing the objective performance. POD⁹ can also extract the main features by decomposing data into a set of orthogonal basis vectors. Duan et al.¹⁰ introduced POD in rotor blade optimization and summarized geometric knowledge from principal component analysis, finding that the knowledge-based design significantly outperformed the reference design. Oyama et al.¹¹ used POD to extract design information from the Pareto front solutions of transonic airfoils, categorizing the optimized shapes into three types: low drag, high lift-to-drag ratio, and high lift, and provided corresponding design guidelines for each type. Secondly, data mining can be used to identify the correlations among multiple design objectives. SOM¹², based on unsupervised neural networks, plays a key role in this task. Song et al.¹³ combined SOM and ANOVA to summarize the influence of design variables on multiple key performance indicators, and validated the accuracy of the knowledge extraction by comparing the shape of optimal solutions obtained from the Metamodel-Based Global Optimization (MBGO) algorithm. Adjei et al.¹⁴ used SOM to capture the correlation between design variables and the performance of two fan rotors. Finally, clustering and other data mining methods can be used to summarize the common characteristics of samples. Guo et al.¹⁵ proposed a knowledge mining framework that integrates information from various data mining methods, such as ANOVA and parallel coordinate plots, and identified optimized design features of a low aspect ratio blade row with endwall/blade body integrated shaping. Qin et al.¹⁶ proposed a Metamodel-Driven Data Mining (MDDM) method that integrates statistical descriptions, SOM, k-d trees, and Sobol indices to perform statistical, correlation, clustering, and sensitivity analyses. They successfully extracted design knowledge and enhanced compressor performance by summarizing design criteria and setting parameters accordingly. Recently, Baert et al.¹⁷ combined surrogate-model-based dynamic optimization with efficient data mining methods to guide high-dimensional design space optimization for low-pressure turbines, achieving an overall efficiency improvement of nearly 0.5%.

In addition to traditional data mining methods, to further clarify the relationship between input design variables and output performance indicators, researchers have recently applied SHapley Additive exPlanations (SHAP) based on cooperative game theory to the field of turbomachinery. This data mining method has also demonstrated advantages in local explanation for individual samples¹⁸⁻²⁰. Shen et al.²¹ conducted multi-objective optimization on a spacecraft example by using LightGBM to build a surrogate model and SHAP explainer to quantify variable contributions. They also proposed the SHAP-shape map method, which visualizes SHAP values on geometric surfaces. Zhang et al.²²

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developed a data mining framework based on metamodel explanation, employing an XGBoost metamodel combined with SHAP explainer, applied to the efficiency and stability enhancement design guidelines mining of a marine gas turbine 3.5-stage transonic axial compressor. The results showed that the compressor stall margin increased to 5.87% at partial rotational speeds after adjusting blade lean and twist, and was further expanded to 23.31% with additional variable stator adjustments. Moreover, the peak isentropic efficiency at the design speed improved by 0.06%. Palar et al.^{23,24} established an interpretable surrogate modeling framework using SHAP and conducted comparative analyses with active subspaces and Sobol indices. The results demonstrated that SHAP offered more informative insights, particularly regarding the directionality of influence exerted by design variables on the objectives, and effectively revealed the intricate trade-offs among multiple performance targets. Wang et al.²⁵ applied the SHAP method to interpret neural network behavior, yielding information on both the sensitivity of individual geometric variables and the coupling effects among multiple geometrical features.

In summary, the application of SHAP—a novel data mining method—to high-dimensional design spaces of turbine stages remains limited in the current literatures^{26,27}. Moreover, most existing analyses focus solely on global scalar performance metrics, rather than high-dimensional vector performance distributions^{28–30}. As a result, current data mining efforts remain constrained to identifying significant variables and their directions of influence, without advancing to explain the underlying mechanisms. To address this gap, this study applies SHAP to explore the 93-dimensional design space of the first stage of the GE-E3 high-pressure turbine. A set of SHAP-based methods is developed to analyze spanwise performance curves and meridional physical field contours. Furthermore, by leveraging the local interpretability of SHAP, attribution analysis of typical samples is performed. Together, these developments form a SHAP-driven data mining framework that provides a comprehensive understanding of how each design variable affects turbine performance and the associated flow mechanisms.

2. SHAP-based data mining framework

2.1. XGBoost-driven SHAP analysis method

SHAP explains the impact of design variables on the final prediction results by attributing contributions to the feature values of individual instances. Based on the dataset generated for the studied problem, predictions are made through model construction, while SHAP values are computed to interpret the model. This study employs a combination of the XGBoost metamodel and the SHAP explainer to analyze the influence of variables on target performance within the black-box design space.

2.1.1. XGBoost metamodel

XGBoost improves model performance and accuracy by executing an ensemble of sequential decision trees. The algorithm iteratively reduces the prediction error of the model, while incorporating a regularization term at each step to prevent overfitting and thus enhance the model's generalization ability. Fundamentally, it constructs a strong ensemble model by combining multiple decision trees to predict the output of the studied problem. The formula is expressed as

$$\hat{y}_i^{(t)} = \sum_{k=1}^t f_k(\mathbf{x}_i) = \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i) \quad (1)$$

where $\hat{y}_i^{(t)}$ represents the model prediction value of the input feature $\mathbf{x}_i$ at the t th iteration, $\hat{y}_i^{(t-1)}$ represents the prediction value retained after the previous $t-1$ iterations, $f_t(\mathbf{x}_i)$ represents the fitted value of the new tree model constructed in the current iteration for the residual of the previous prediction.

In the iterative process, the optimization objective of XGBoost consists of two parts: the loss function and the regularization term, expressed as

$$\text{Obj} = \sum_i l(y_i, \hat{y}_i) + \Omega(f_k) \quad (2)$$

where $l(y_i, \hat{y}_i)$ represents the loss function, measuring the discrepancy between the true values and the predicted values, thus reflecting the model's accuracy; $\Omega(f_k)$ denotes the regularization term, which accounts for the model's complexity. By minimizing the objective function Obj, the newly introduced tree model can be determined.

2.1.2. SHAP explainer

SHAP assigns an “importance” value, called the SHAP value, to each feature by considering all possible combinations of features and their contributions to the prediction. Suppose the i th sample of the studied problem is $\mathbf{x}_i$, the j th feature of this sample is x_{ij} , the model's prediction for this sample is y_i , and the model's baseline value (typically the mean of the target variable across all samples) is y_{base} . Then, the SHAP values satisfy the equation

$$y_i = y_{\text{base}} + f(x_{i1}) + f(x_{i2}) + \dots + f(x_{ik}) \quad (3)$$

where $f(x_{ij})$ is the SHAP value of x_{ij} . Intuitively, $f(x_{i1})$ represents the contribution of the 1th feature in the i th sample to the final prediction y_i . When $f(x_{i1}) > 0$, it indicates that this feature increases the prediction value, exerting a positive effect; conversely, if $f(x_{i1}) < 0$, it means the feature decreases the prediction value, having a negative effect.

SHAP is a sample-based local explanation method, whose greatest advantage lies in its ability to reflect the impact of each feature in every individual sample through SHAP values. Calculating SHAP values involves computing the marginal contribution of each feature, which can be simply understood as the difference between the contribution when a certain feature is included and the contribution before that feature is added. The specific calculation method is as follows

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} [f_x(S \cup \{i\}) - f_x(S)] \quad (4)$$

where $N = \{1, 2, \dots, M\}$ represents the index set of feature variables in the dataset, and M is the total number of feature variables; S is a subset of $\{1, 2, \dots, i-1, i+1, \dots, M\}$, and $|S|$ is the number of elements in subset S ; $f_x(S \cup \{i\})$ denotes the model's prediction when only the features in $S \cup \{i\}$ are used, and $f_x(S)$ denotes the model's prediction when only the features in S are used. The difference between them represents the marginal contribution of the i th feature under subset S .

2.2. Data mining framework based on SHAP

Based on the SHAP data mining method, a data mining framework is established in this study, as illustrated in Fig. 1. After collecting sample data in the design space, significant variables are first identified through bar chart constructed using SHAP values. On this basis, the influence of these variables on overall performance (0-dimension data) is summarized. Subsequently, the response of spanwise performance (1-dimension data) is analyzed to determine which regions of the spanwise are affected by the variables to improve overall performance. Furthermore, the physical mechanisms by which variables affect performance are examined using meridional physical field contours (2-dimension data). Finally, leveraging the local interpretability of SHAP, typical samples in the design space are selected for attribution analysis.

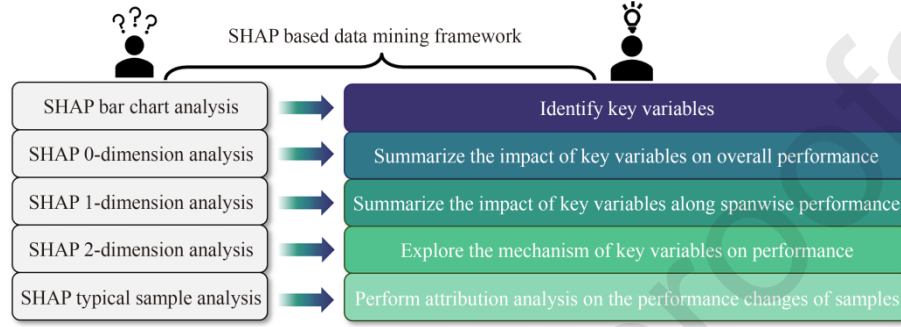


Fig. 1 SHAP based data mining framework

3. Research object and design space

3.1. Research object and numerical method

Fig. 2 presents the geometric configuration of the first stage of the GE-E3 High-Pressure Turbine (HPT), which serves as the research object in this study. The geometry consists of four components: stator (S), rotor (R), hub surface, and casing surface. The stator consists of 46 blades, while the rotor contains 76 blades. The rotor tip clearance is 0.3 mm.

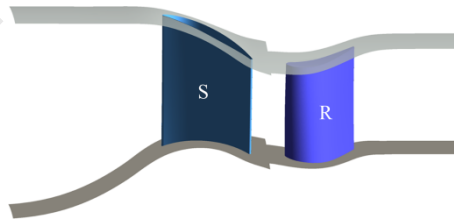


Fig. 2 Geometric diagram of first stage of GE-E3 HPT

The commercial software NUMECA was employed to perform mesh generation and aerodynamic performance simulations. The Spalart–Allmaras turbulence model was adopted. The turbine stage was evaluated under the following operating conditions: inlet total temperature of 719 K, inlet total pressure of 344.74 kPa, outlet static pressure of 139.473 kPa, and rotor rotational speed of 8279 r/min. A grid independence study and experimental validation were conducted, with the results presented in Figs. 3 and 4. The grid independence analysis shows that both efficiency and pressure ratio stabilize when the grid count exceeds approximately 1.5 million nodes. Considering both computational accuracy and resource consumption, a structured grid scheme with about 2 million nodes was ultimately selected for this study. Fig. 4 presents a comparison between the numerically computed isentropic Mach number distribution on the stator blade surface and experimental data from the literature³¹. The numerical results agree well with the

experimental measurements in both overall trend and magnitude, demonstrating good consistency. The above comparisons verify the accuracy of the numerical method employed in this study. The total-to-total efficiency (hereinafter referred to as “efficiency”) was selected as the key indicator to evaluate the aerodynamic performance of the turbine stage, and its expression is given by

$$\eta = \frac{1 - T_2^* / T_0^*}{1 - (p_2^* / p_0^*)^{(k-1)/k}} \quad (5)$$

where subscripts 0 and 2 denote the inlet and outlet of the turbine stage, respectively; T^* and P^* represent the total temperature and total pressure; and k is the specific heat ratio. Considering that the turbine stage must operate close to its design point, the variation in mass flow rate is also monitored in addition to the efficiency.

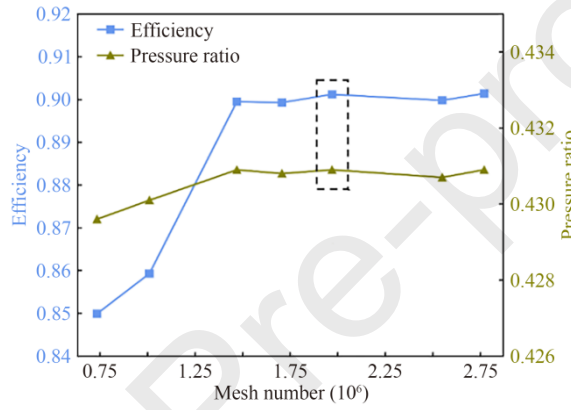


Fig. 3 Grid independence verification

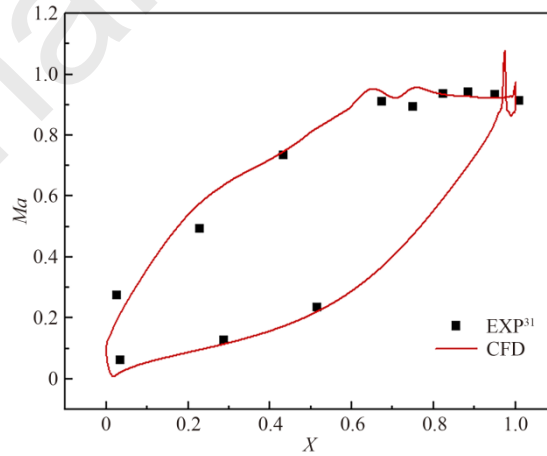


Fig. 4 Experimental validation

3.2. Geometric parameterization method and design space

The blade shaping adopts a geometric parameterization method. For the stator, three cross-sectional profiles—hub,

mid-span, and tip—are generated using 45 control parameters, with 15 parameters defined for each section. Additionally, three parameters are employed to adjust the three-dimensional stacking shape of the stator, the specific definitions of the parameters are shown in Table 1. The control parameters are illustrated in Fig. 5, where Fig. 5(a) marks the sectional profile parameters, and Fig. 5(b) depicts the variation scheme of the three-dimensional stacking parameters. The radial stacking line is fixed at the hub, while control points are placed at the mid-span and tip positions. This allows the stacking line to shift circumferentially at the mid-span and to shift in both circumferential and axial directions at the tip, thereby achieving three-dimensional lean and sweep shaping. For the rotor, considering structural strength requirements during rotation, only the 45 sectional profile parameters are adjusted, without modifying the three-dimensional stacking shape. Based on this method, a total of 93 parameters are used to precisely describe the geometry of both blade rows, forming a 93-dimensional design space. For clarity, S1, S2, and S3 represent the hub, mid-span, and tip sections of the stator, respectively; while R1, R2, and R3 correspond to the respective sections of the rotor. D_1 , D_2 , and D_3 denote the circumferential mid-span shift, circumferential tip shift, and axial tip shift of the stator, respectively.

The value range of each design variable was specified within the design space. Subsequently, the Latin Hypercube Sampling (LHS) method was employed to generate the sample set. All design variables were normalized to the range of (0, 1) based on the reference blade geometry, where a value of 0.5 corresponds to the baseline.

Table 1 Specific parameter definition of geometric parameter method

No.	Design parameter	Symbol	Description
1	Leading edge radius	R_L	Radius of inlet circle
2	Axial chord length	L_C	Maximum distance in axial direction
3	Centerline angle between circles	A_C	Angle between centerline of leading-edge and trailing-edge circles and axial direction
4	Inlet geometric angle	A_I	Angle between inlet direction and leading edge line
5	Leading edge correction	C_L	Parameter correcting the suction side leading edge
6	Effective outlet angle	A_E	$\arcsin(o/t)$, where o throat diameter, t is pitch
7	Outlet deflection angle	A_D	Angle between outlet circle tangent and suction side trailing edge tangent
8	Correlation coefficient	C_C	Controls the uncovered curve segment at the throat outlet
9	Outlet wedge angle	A_W	Angle between suction side trailing edge and pressure side trailing edge tangent
10	Displacement of control point 1 at suction side leading edge	P_{SL1}	Displacement of Bezier spline control point 1 of suction side first curve segment
11	Displacement of control point 2 at suction side leading edge	P_{SL2}	Displacement of Bezier spline control point 2 of suction side first curve segment
12	Displacement of control point 1 at suction side trailing edge	P_{ST1}	Displacement of Bezier spline control point 1 of suction side second curve segment
13	Displacement of control point 2 at suction side trailing edge	P_{ST2}	Displacement of Bezier spline control point 2 of suction side second curve segment
14	Displacement of control point 1 at pressure side	P_{P1}	Displacement of Bezier spline control point 1 of pressure side curve
15	Displacement of control point 2 at pressure side	P_{P2}	Displacement of Bezier spline control point 2 of pressure side curve
16	Three-dimensional stacking parameters	$D(D_1, D_2, D_3)$	Stator circumferential middle shift, circumferential top shift, axial top shift

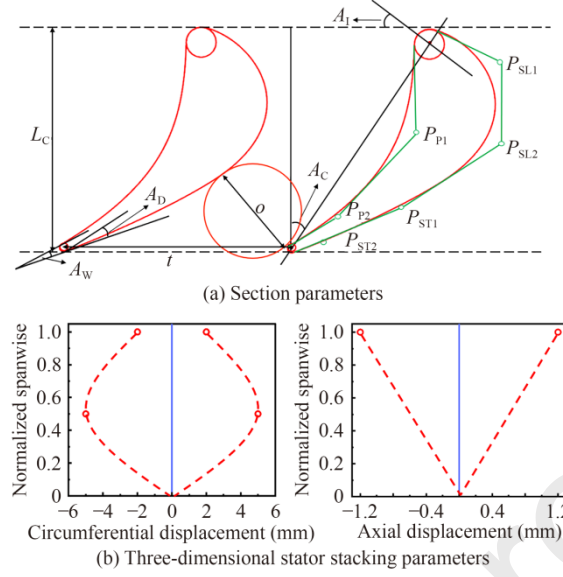


Fig. 5 Control parameters diagram

4. Data mining and mechanism analysis of high-pressure turbine stage

4.1. Dataset preparation

Based on Latin Hypercube Sampling (LHS) technique, 1000 samples were generated from the aforementioned parameterized design space for preliminary exploration. Fig. 6 presents boxplot illustrating the distributions of the output variables—efficiency and mass flow rate—along with their respective baseline values. The means and medians of both variables are very close, indicating approximately normal distributions. For efficiency, nearly two-thirds of the samples exhibit values exceeding the baseline reference design, suggesting a significant potential for efficiency improvement within the design space. The maximum sampled efficiency reaches 91.26%. Regarding mass flow rate, most samples also surpass the baseline value, indicating a tendency for increased mass flow rate in the design space. The difference between the maximum and minimum flow rates is 1.11 kg/s, corresponding to a variation magnitude of 9.6% relative to the reference design, which represents a relatively large range.

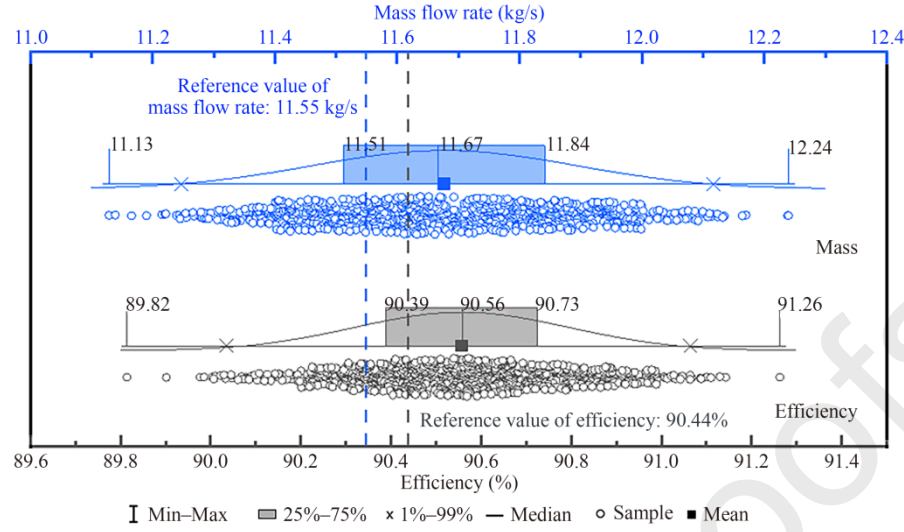


Fig. 6 Boxplot of key performance indicators

Based on the above statistical analysis, in order to prevent significant increases in stage mass flow rate and deviations from the design operating condition during the turbine stage design process, an additional constraint was introduced to retain only those samples whose mass flow rate deviates by no more than $\pm 1\%$. This filtering resulted in a reduced 93-dimensional design space dataset containing 352 samples, which was used for subsequent analysis.

4.2. Significant variables analysis

Based on the 352 samples within the design space, a sensitivity analysis was first conducted. The XGBoost algorithm was employed to construct the surrogate model, and its prediction accuracy is shown in Fig. 7. It can be observed that the coordinate points formed by the predicted and true efficiency values are densely distributed along the 45° line, the calculated root mean square error for the cross-validation was 0.0290%, and the relative error was 0.0274%, indicating that the constructed surrogate model exhibits high predictive accuracy for turbine efficiency.

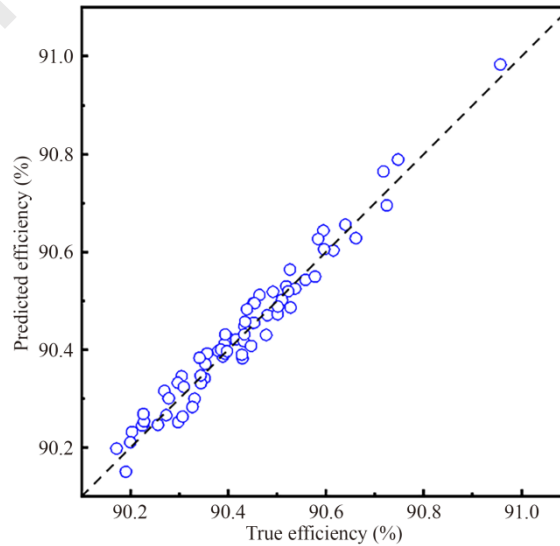


Fig. 7 Precision check of approximate model of efficiency

SHAP values were calculated using the SHAP explainer to identify the significant variables. Design variables corresponding to the same geometric parameter type were grouped and ranked accordingly. The results are presented as a SHAP bar chart in Fig. 8. With respect to stage efficiency, the top three most influential parameters are the blade effective outlet angle (A_E), the stator three-dimensional stacking parameters (D), and the spline control point near the leading edge of the suction side (P_{SL2}). In addition, among the six blade sections of the stator and rotor, the middle section of the rotor (R2) and that of the stator (S2) exhibit a higher level of influence on performance metrics compared to other sections.

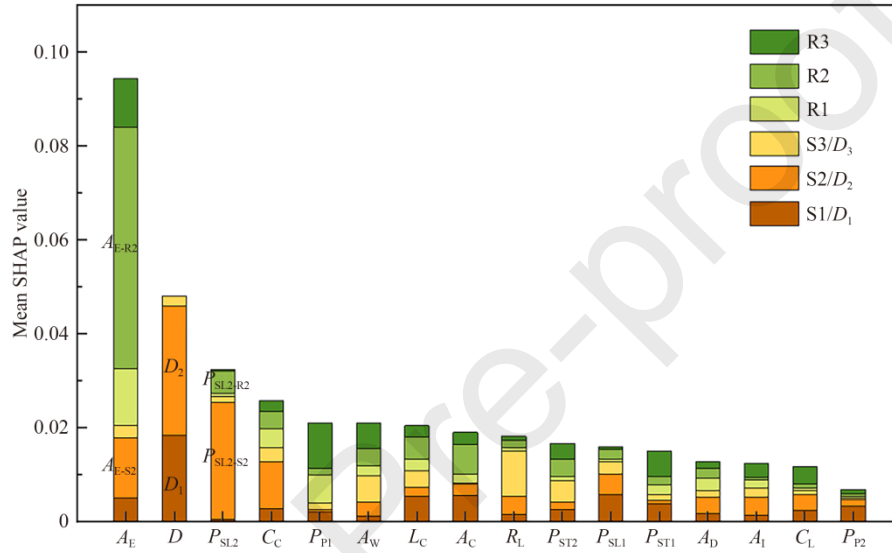


Fig. 8 SHAP bar chart significant variable recognition

The parameters A_E and P_{SL2} , ranked 1st and 3rd respectively, belong to the primitive shaping category, while the parameter D , ranked 2nd, belongs to the stacking shaping category. The subsequent analysis will focus on the top-ranked design variables within each of these three parameter categories (the six variables annotated in Fig. 6), and conduct further investigation using SHAP-based 0-dimension, 1-dimension, and 2-dimension analysis methods. The goal is to explore how these variables influence the overall performance of the turbine stage and the spanwise performance distribution, along with the underlying mechanisms.

4.2.1. Effective outlet angle

As the geometric parameter with the greatest impact on stage efficiency, the effective outlet angle mainly influences performance by altering the throat geometry of the flow passage. Fig. 9 illustrates the effect directions of the effective outlet angle at the rotor middle section and stator middle section on turbine stage efficiency. Each point in the figure represents a sample, with the color indicating the value of the design variable. The vertical axis lists the variable names, while the horizontal axis shows the SHAP values. A positive SHAP value indicates a positive influence of the input parameter on the performance metric, whereas a negative SHAP value indicates a negative influence. Additionally, the design variables are ranked according to their feature importance. It can be observed that a smaller effective outlet angle at the rotor middle section (A_{E-R2}) corresponds to a positive effect on output performance, i.e., higher efficiency; conversely, a smaller effective outlet angle at the stator middle section (A_{E-S2}) corresponds to a negative effect on output performance, i.e., lower efficiency. In other words, to improve turbine stage efficiency, the effective outlet angle at the rotor middle section should be decreased, while that at the stator middle section should be increased.

Previous data mining studies often focused on the analysis of overall turbine stage performance (0-dimension data). In contrast, this work develops an analysis method for the performance distribution along the spanwise direction of the turbine stage (1-dimension data), enabling a more intuitive demonstration of how design variables improve overall performance by affecting specific spanwise locations. Fig. 10(a) shows the influence of the effective outlet angle at the rotor middle section on the efficiency distribution along the spanwise direction. The vertical axis represents the dimensionless spanwise position, while the horizontal axis shows the effect direction on efficiency via SHAP values. Using 1-dimension SHAP visualization, it is clearly seen that decreasing A_{E-R2} increases efficiency over a wide mid-span region, thereby improving overall performance. However, the decrease of A_{E-R2} also causes efficiency reduction in the upper endwall region. Fig. 10(b) depicts the effect of the effective outlet angle at the stator middle section on the efficiency distribution along the spanwise direction. Increasing A_{E-S2} can enhance efficiency over most of the spanwise range by increasing the stage flow rate, although efficiency slightly decreases near both the upper and lower endwall regions.

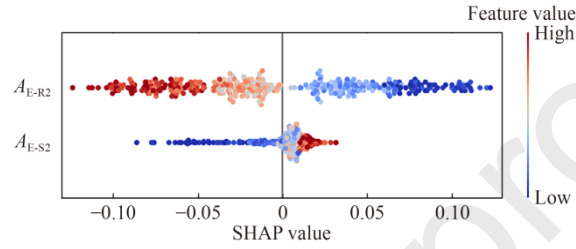


Fig. 9 Effect of A_E on overall efficiency of turbine stage

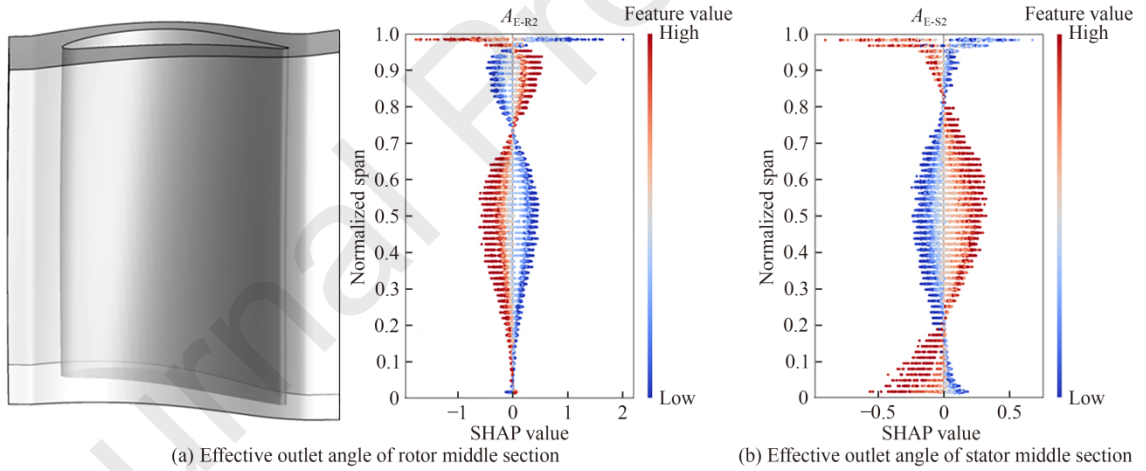


Fig. 10 Effect of A_E on turbine efficiency along spanwise

Further mechanistic explanations are made through data mining on the meridional plane physical field contour plots (2-dimension data). SHAP values of the concerned performance metrics are calculated at 128×64 data points on the meridional plane. For each variable, the mean absolute SHAP values across all samples at each data point are used to plot the sensitivity map of the performance metric with respect to that variable; meanwhile, the variable's value range from 0 to 1 is divided into 10 intervals with 0.1 steps. The mean SHAP values within each interval are plotted at each data point to show the trend of the performance with the variable change. Fig. 11 shows the entropy response on the meridional plane to the effective outlet angle at the rotor mid-span. From the sensitivity analysis in Fig. 11(a), A_{E-R2} mainly affects the entropy values at the rotor outlet, the mid-span region of the downstream passage, and the upper

endwall area. Trend analysis in Fig. 11(b) reveals that with decreasing A_{E-R2} , entropy decreases and efficiency increases over the mid-span region, whereas entropy increases and efficiency decreases near the upper endwall. This agrees with the conclusions from the spanwise performance variation. This is because for the rotor, decreasing the mid-span effective outlet angle increases rotor reaction degree, which reduces profile loss and entropy in the mid-span area; however, secondary flow loss increases near the endwall, causing higher entropy there. Fig. 12 shows the entropy response on the meridional plane to the effective outlet angle at the stator mid-span. Sensitivity analysis in Fig. 12(a) indicates that A_{E-S2} mainly influences the stator region, especially entropy near the stator outlet, and also affects the entropy near the upper and lower endwalls downstream. Trend analysis in Fig. 12(b) further shows that increasing A_{E-S2} leads to higher entropy because A_{E-S2} increase can raise stage mass flow within the constraint limits, improving efficiency; however, increased flow leads to higher losses and entropy, especially near the endwalls. It should be noted that samples with flow variation beyond $\pm 1\%$ have been filtered out previously. Since A_{E-S2} significantly affects stage flow, all samples in the $[0.9, 1.0]$ range for A_{E-S2} were excluded, thus no SHAP values appear in the contour plots for this range.

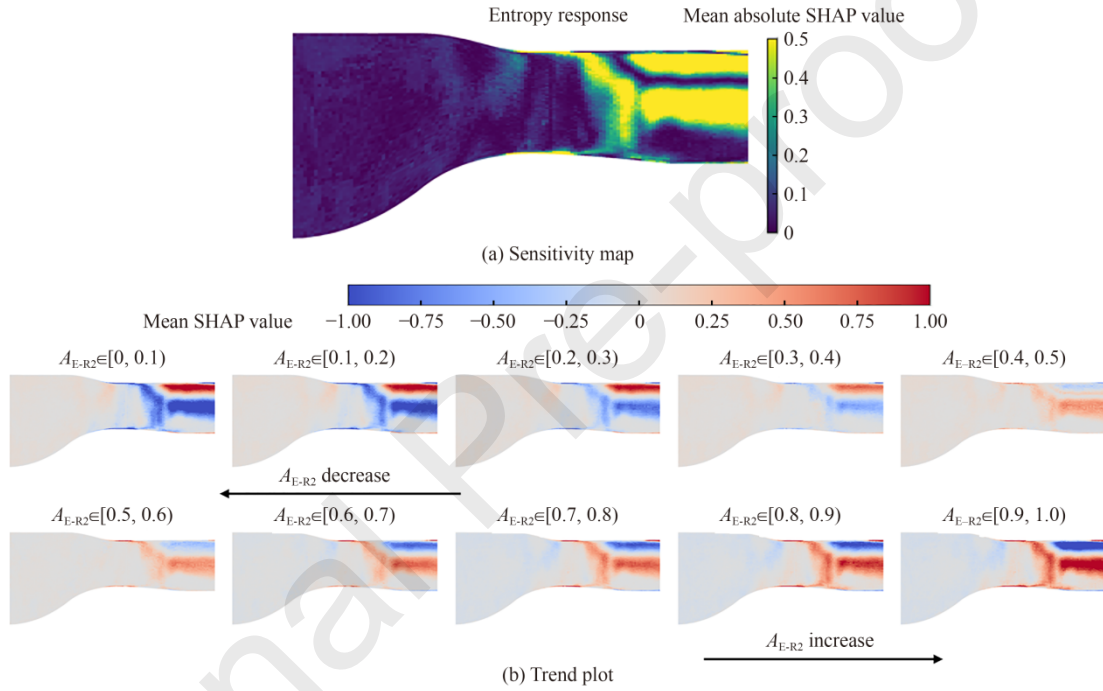


Fig. 11 Effect of A_{E-R2} on distribution of entropy on meridional plane of turbine

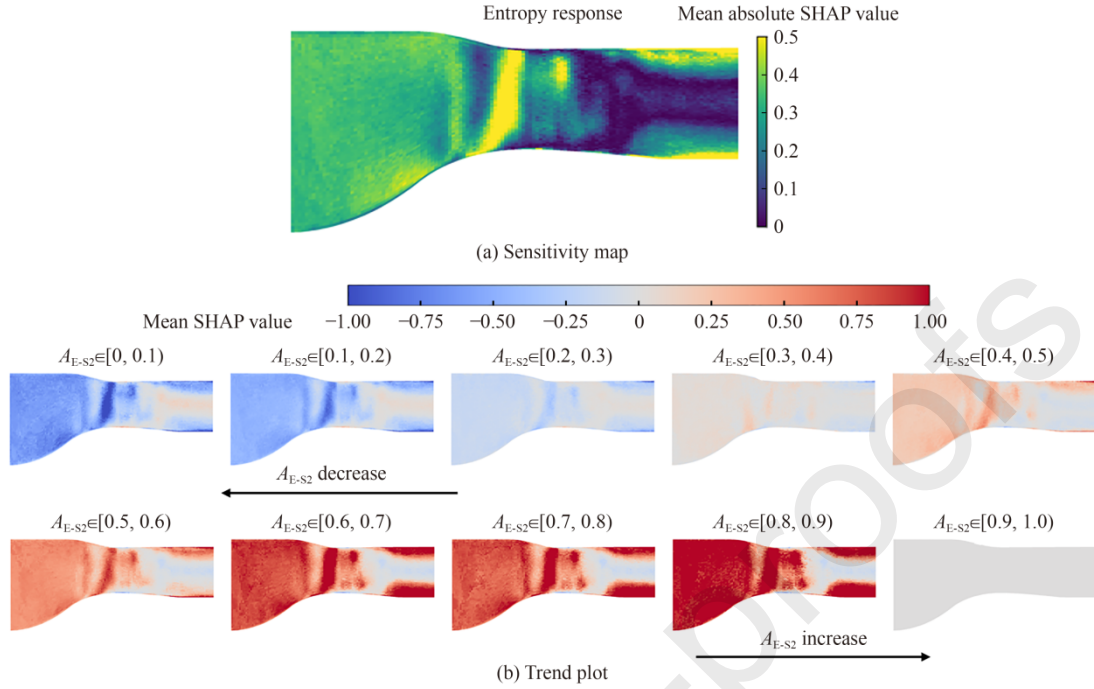


Fig. 12 Effect of A_{E-S2} on distribution of entropy on meridian plane of turbine

Since the effective outlet angle alters the throat geometry and consequently affects the turbine flow capacity, SHAP scatter plots are further employed to simultaneously illustrate the directional influence of varying effective outlet angles on turbine stage efficiency and mass flow. The correlation between these two performance indicators is investigated, as shown in Fig. 13. In the plots, the color of each scatter point denotes the magnitude of the design variable, the vertical axis indicates its contribution to stage efficiency, and the horizontal axis represents its contribution to stage mass flow. Fig. 13(a) presents the trend associated with the rotor mid-span effective outlet angle. As A_{E-R2} decreases, the turbine stage efficiency increases while the mass flow decreases. Fig. 13(b) shows the trend corresponding to the stator mid-span effective outlet angle. As A_{E-S2} increases, both the stage efficiency and the mass flow increase. In turbine stage design, it is typically desirable to maximize efficiency without altering the design operating condition, i.e., maintaining a consistent mass flow rate. The combined effect of decreasing the rotor mid-span effective outlet angle and increasing the stator mid-span effective outlet angle allows for efficiency enhancement while constraining variations in mass flow, representing a synergistic design approach where the overall benefit exceeds the sum of individual effects.

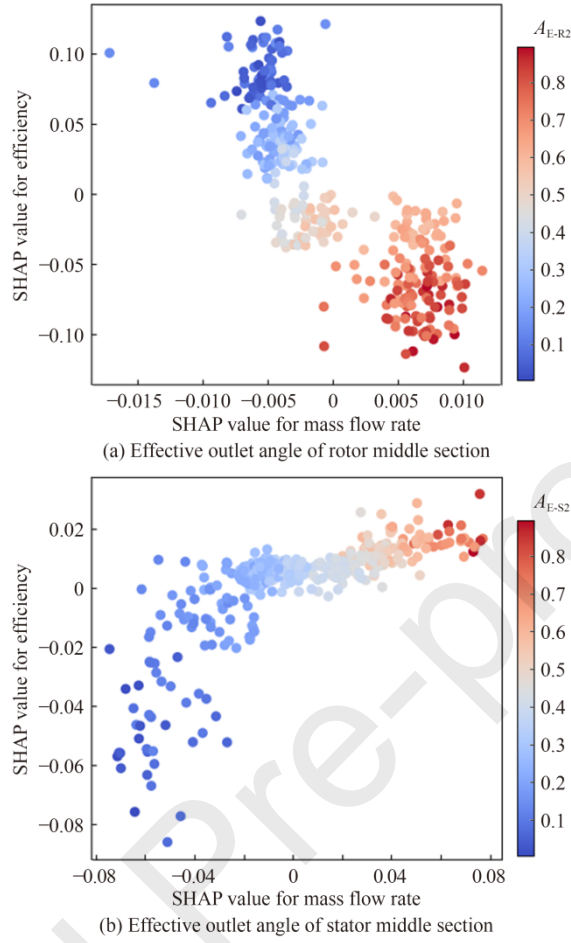


Fig. 13 Turbine stage efficiency and mass flow rate correlation

4.2.2. Three-dimensional stacking parameters of stator

The three-dimensional stacking parameters of the stator are also important factors affecting turbine stage efficiency. Three-dimensional design techniques such as blade sweep and lean can improve stage efficiency by altering the radial pressure gradient within the blade row. Fig. 14 presents the effects of two significant variables among the three-dimensional stacking parameters—circumferential tip shift D_2 and circumferential mid-span shift D_1 —on efficiency. It is observed that both variables contribute to improved turbine stage efficiency when taking negative values.

The above phenomena are further interpreted through 1-dimension SHAP analysis. Fig. 15 illustrates the effects of the stator three-dimensional stacking parameters on the efficiency distribution along the spanwise direction. When D_2 takes a negative value, i.e., the circumferential tip shift moves toward the pressure side of the blade, it enhances the efficiency in the tip region. When D_1 takes a negative value, i.e., the circumferential mid-span shift moves toward the pressure side, it improves the efficiency in the mid-span region. The combined effect of both shifts results in a positively leaned blade, which optimizes the radial pressure gradient within the blade passage and contributes to enhanced stage efficiency.

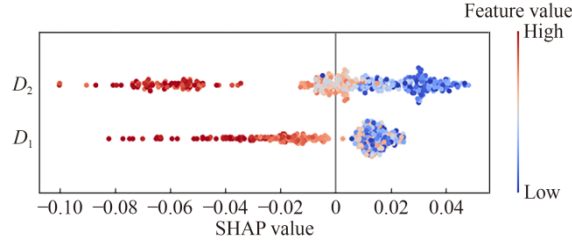
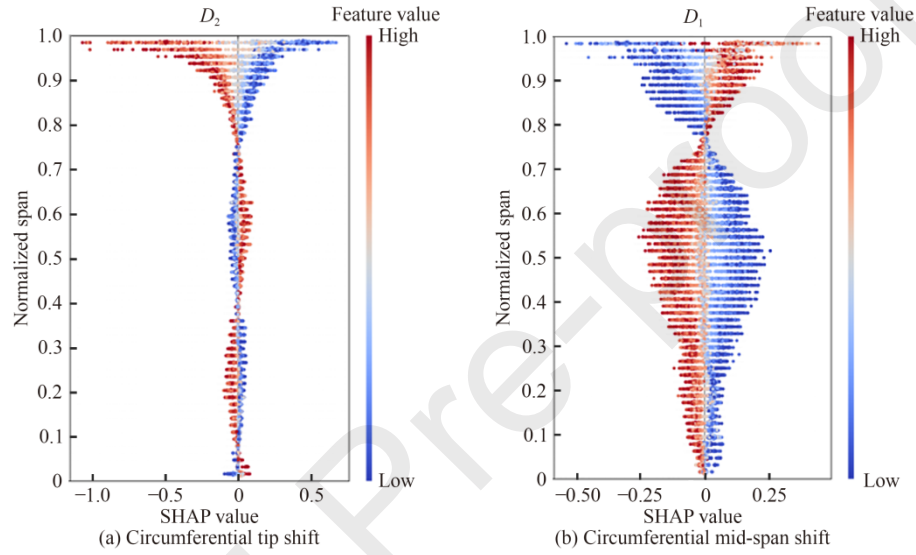
Fig. 14 Effect of D on overall efficiency of turbine stageFig. 15 Effect of D on turbine efficiency along spanwise

Fig. 16 shows the response of meridional static pressure to the circumferential tip shift. As observed in the sensitivity analysis of Fig. 16(a), D_2 significantly affects the static pressure in the tip region of the stator blade. Furthermore, the trend in Fig. 16(b) reveals that when the tip shift takes a negative value (i.e., shifts toward the pressure side), the static pressure in the tip region decreases, thereby improving the radial pressure gradient in this area. Fig. 17 presents the response of meridional static pressure to the circumferential mid-span shift. As shown in Fig. 17(a), D_1 significantly affects the static pressure over a wide range in the mid-span and upper endwall regions, and to a lesser extent in the lower endwall region. As illustrated in Fig. 17(b), a negative mid-span shift (i.e., toward the pressure side) results in a “C”-shaped radial static pressure distribution, with higher values at the two ends and lower values in the middle region. The reduced static pressure in the middle region intensifies the radial pressure gradient near the tip, leading to reduced efficiency, while the improved pressure gradient in the mid-lower regions enhances efficiency.

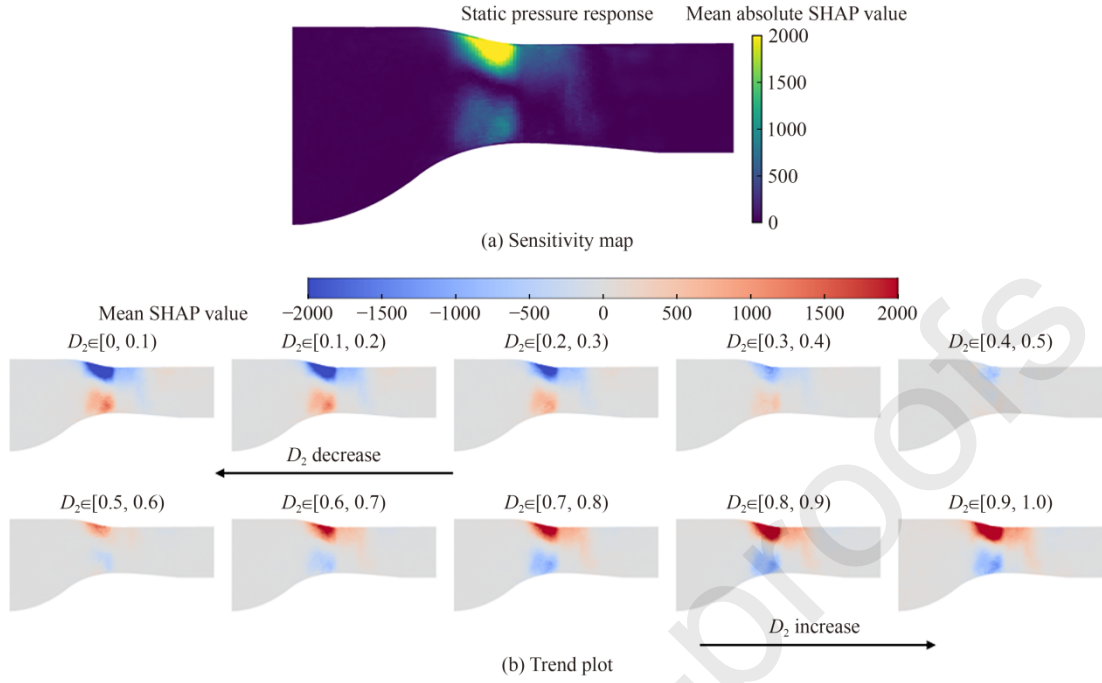


Fig. 16 Effect of D_2 on distribution of static pressure on meridian plane of turbine

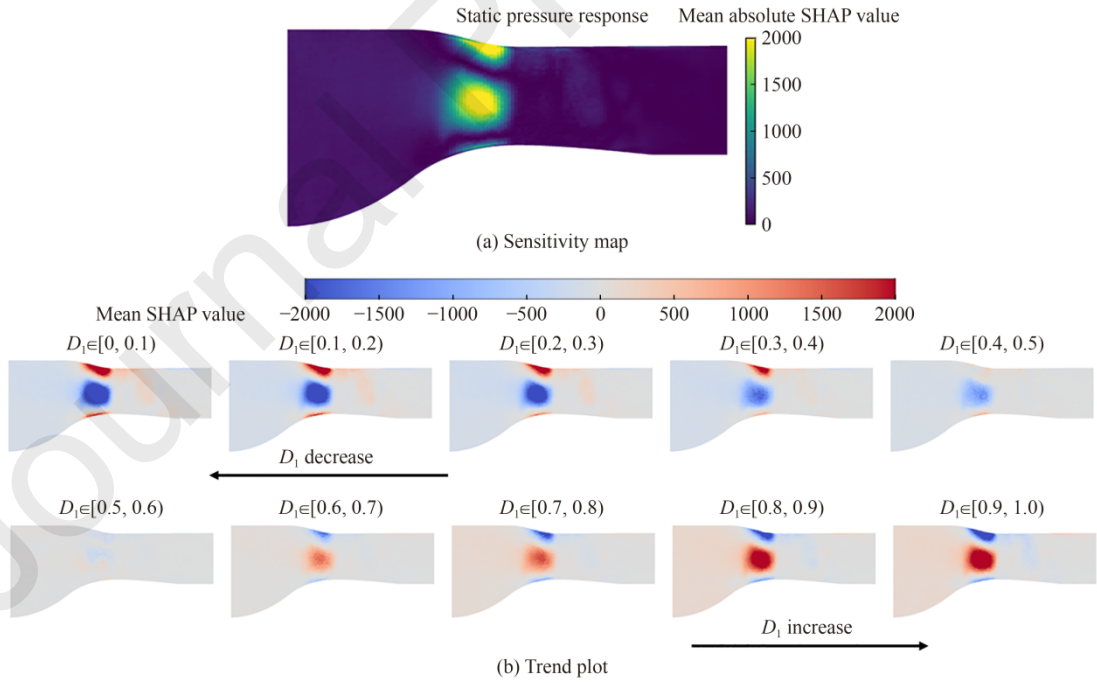


Fig. 17 Effect of D_1 on distribution of static pressure on meridian plane of turbine

4.2.3. Spline control points on suction-side leading edge of blade—SLE2

The spline control points near the suction-side leading edge SLE2 at the mid-span sections of stator and rotor blades also affect turbine stage efficiency, as illustrated in Fig. 18. Simultaneous reductions in P_{SL2-S2} and P_{SL2-R2} contribute to improvements in turbine stage efficiency.

Fig. 19 focuses on the efficiency improvement regions. A reduction in the value of the control points near the suction-side leading edge at the mid-span section, i.e., shifting toward a thinner leading-edge shape, helps reduce the pressure gradient in the leading-edge acceleration zone. This leads to lower irreversible flow losses and improved turbine stage efficiency across most of the spanwise direction. By comparing Fig. 19(a) and Fig. 19(b), it can be observed that the impact of P_{SL2} at the rotor mid-span section is relatively smaller than that at the stator mid-span. Additionally, the presence of tip clearance causes a certain influence on the rotor tip efficiency.

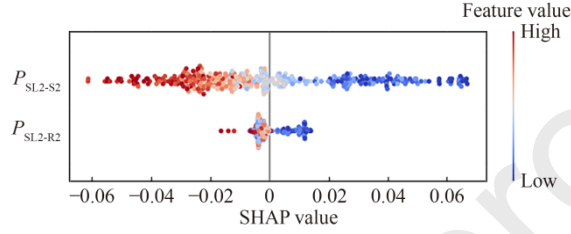


Fig. 18 Effect of P_{SL2} on overall efficiency of turbine stage

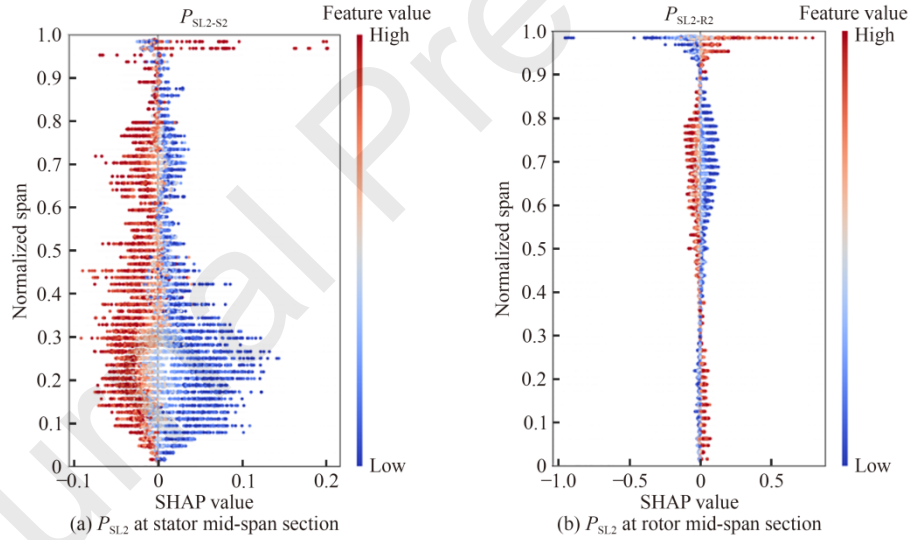


Fig. 19 Effect of P_{SL2} on turbine efficiency along spanwise

Figs. 20 and 21 provide further explanations of the underlying mechanisms. Fig. 20 shows the response of the meridional mach number to the suction-side leading-edge control point P_{SL2} at the stator mid-span section. As shown in Fig. 20(a), under the influence of the radial equilibrium equation, P_{SL2-S2} affects the mach number distribution across the entire stator region. When the value of P_{SL2-S2} decreases, the mach number in the leading-edge acceleration region decreases. This confirms the conclusion from the spanwise performance analysis that the irreversible flow losses in this region are reduced and the efficiency is improved. Fig. 21 illustrates the response of the meridional mach number to the suction-side leading-edge control point P_{SL2} at the rotor mid-span section. The influence of P_{SL2-R2} on the mach number is consistent with that of P_{SL2-S2} and will not be further elaborated. It is noteworthy that P_{SL2-R2} also has a significant impact on the mach number at the rotor inlet.

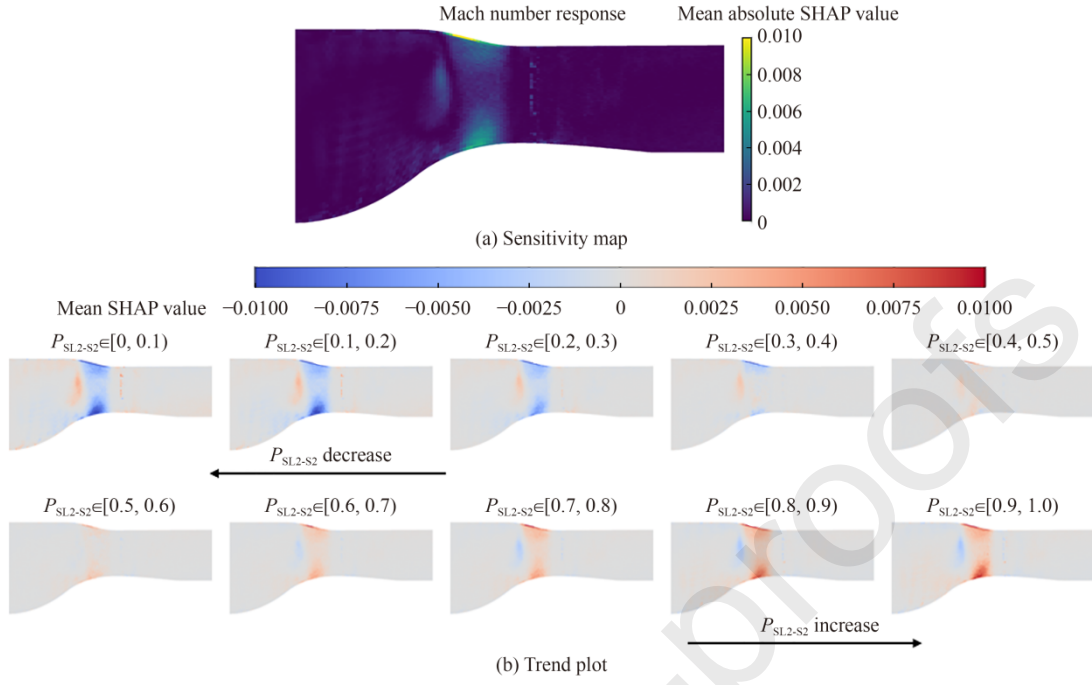


Fig. 20 Effect of P_{SL2-S2} on distribution of mach number on meridian plane of turbine

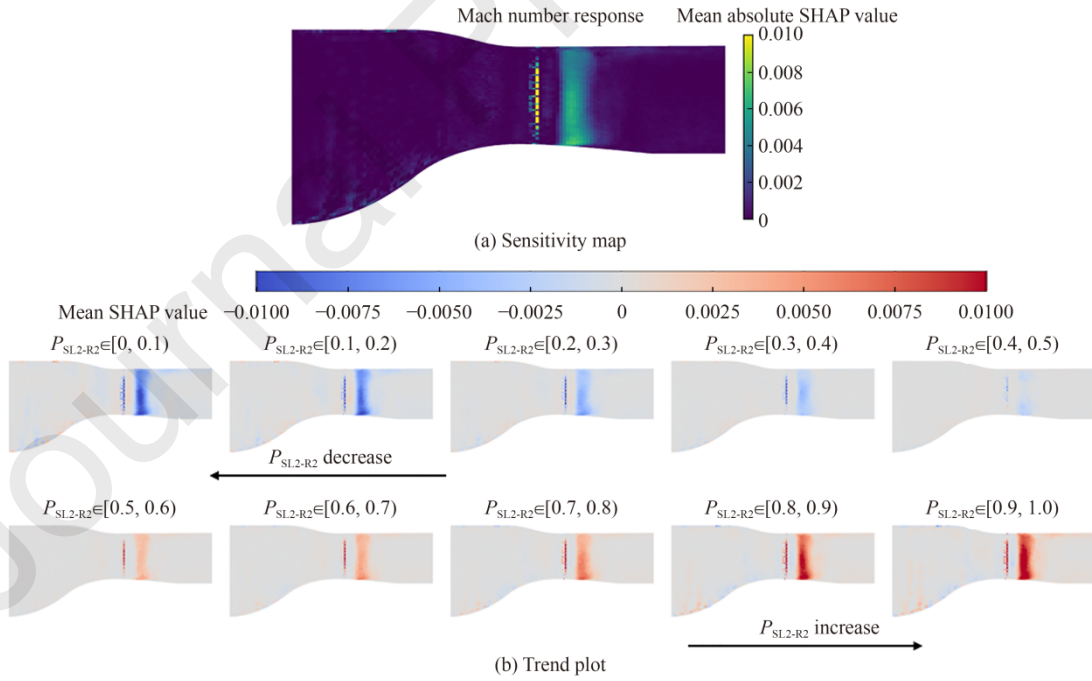


Fig. 21 Effect of P_{SL2-R2} on distribution of Mach number on meridian plane of turbine

4.3. Attribution analysis of typical samples

In addition to the analysis from 0-dimension to 1-dimension data, a major advantage of SHAP over other globally interpretable data mining methods lies in its local interpretability. SHAP enables attribution analysis at the level of individual samples, allowing for the visualization of the role each design variable plays in the variation of the performance indicator for a specific sample.

The typical samples with the maximum and minimum efficiency in the trimmed dataset were selected for analysis, as shown in Fig. 22. Fig. 22(a) shows the sample with the maximum efficiency. Compared with the average efficiency ($E[f(x)]$) of 90.456% across the dataset, this sample achieved an increased efficiency ($f(x)$) of 90.956%. Key variables contributing to this improvement include the effective outlet angle at the mid-span of the rotor (A_{E-R2}), the axial chord length at the hub section of the rotor (L_{C-R1}), and the suction-side leading edge spline control point at the mid-span of the stator (P_{SL2-S2}). Specifically, A_{E-R2} has a value of 0.112, which is lower than the baseline value of 0.5 and contributes to a 0.11% increase in efficiency. L_{C-R1} has a value of 0.007, contributing 0.09%, and P_{SL2-S2} has a value of 0.029, contributing 0.05%. Fig. 22(b) presents the sample with the minimum efficiency, which decreases to 90.007% compared with the baseline of 90.456%. The effective outlet angle at the mid-span of the rotor A_{E-R2} takes a value of 0.822, exceeding the baseline of 0.5, leading to a 0.09% decrease in efficiency. The axial chord length at the tip section of the stator L_{C-S3} is 0.007, contributing to a 0.08% decrease, and the circumferential mid-span shift of the stator D_1 is 0.911, leading to a 0.07% decrease.

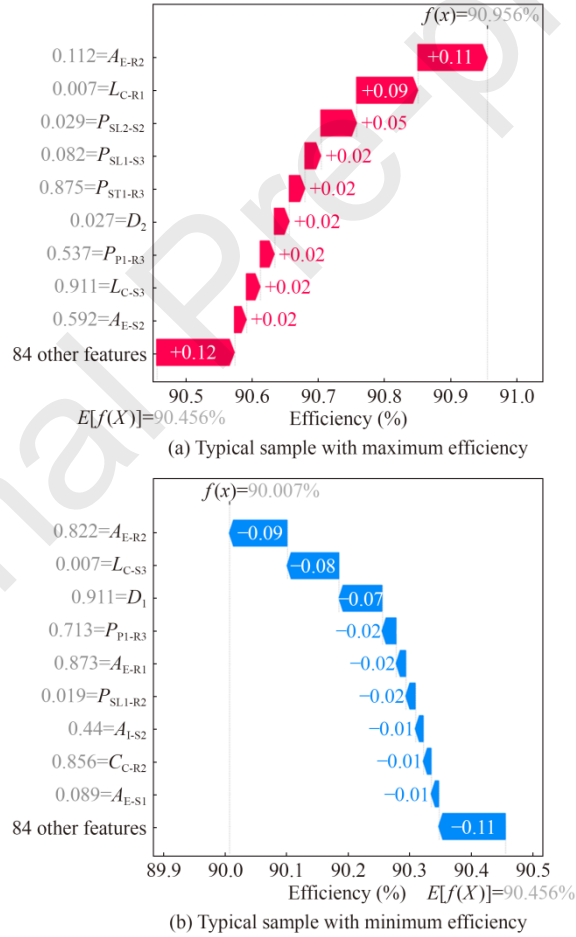


Fig. 22 Typical samples attribution analysis

The attribution analysis of the typical samples with maximum and minimum efficiency reveals that the significant parameters affecting sample performance include the effective outlet angle, suction-side leading-edge control point, stator three-dimensional stacking, and axial chord length. These findings are consistent with the previously identified

significant variables, and the direction of their influence also agrees with the data mining results.

4.4. Design guidelines summary and aerodynamic verification

Based on the above analysis, the design guidelines for improving the stage efficiency of the first-stage high-pressure turbine of GE-E3 can be summarized as follows:

(1) Effective outlet angle: Decrease the effective outlet angle at the mid-span of the rotor, and increase the effective outlet angle at the mid-span of the stator.

(2) Stator three-dimensional stacking parameters: Apply negative shift values at both the circumferential tip and mid-span positions, i.e., shift toward the pressure side.

(3) Suction-side leading-edge control points: Move the control points near the suction-side leading edge at the mid-span of both stator and rotor blades toward the blade thinning direction.

Following the above design guidelines, geometric modifications were carried out in directions beneficial to improving stage efficiency. First, the effective outlet angle, stator three-dimensional stacking parameters, and suction-side leading-edge control points were individually adjusted to create new designs. Then, all the above parameters were modified simultaneously to obtain the final design. A comparison of efficiency values among different designs is presented in Table 2.

Table 2 Comparison of efficiency values among different designs

Geometric design	Efficiency (%)	Improvement (%)
Reference design	90.44	
New design with modified effective outlet angles	90.96	0.52
New design with modified stator three-dimensional stacking parameters	90.58	0.14
New design with modified suction-side leading-edge control points	90.48	0.04
Final design (all modifications applied)	91.09	0.65

As shown in Table 2, all four geometries derived from the design guidelines lead to improvements in turbine stage efficiency. Among the cases where only a single parameter is modified, the effective outlet angle contributes the most to efficiency improvement, followed by the stator three-dimensional stacking parameters, and finally the suction-side leading-edge control points. This ranking is consistent with the importance order of significant variables identified in the SHAP bar chart analysis. The final design, which fully adheres to the design guidelines, achieves a turbine stage efficiency of 91.09%, exceeding the maximum value (90.96%) in the trimmed dataset. This strongly demonstrates the effectiveness and potential of the SHAP-based data mining approach.

CFD simulations are conducted for both the reference design and the final design to further compare the changes in aerodynamic performance of the turbine stage. The entropy function on the meridional plane is used to characterize the irreversible losses in the flow process. It is defined as the entropy produced due to the interaction of viscous effects and velocity gradients in the fluid, and the formulation is

$$\Phi = \mu \left[2 \left(\frac{\partial u}{\partial x} \right)^2 + 2 \left(\frac{\partial v}{\partial y} \right)^2 + 2 \left(\frac{\partial w}{\partial z} \right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right)^2 \right] \quad (6)$$

where μ denotes the dynamic viscosity. The term $2\left(\frac{\partial u}{\partial x}\right)^2, 2\left(\frac{\partial v}{\partial y}\right)^2, 2\left(\frac{\partial w}{\partial z}\right)^2$ represents the stretching or compressing effects of the fluid in various directions, while the term $\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)^2, \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)^2, \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right)^2$ describes the shearing effects between velocity gradients.

Fig. 23 shows the distribution contours of meridional entropy for the reference design and the final design. It can be observed that compared to the reference design, the entropy in the meridional plane is significantly reduced in the final design. In Fig. 23: Region A corresponds to the leading edge of the stator, where the high-entropy regions near the upper and lower endwalls shift downstream in the final design; Region B refers to the interface between the stator and rotor, where the low-entropy region is enlarged in the final design; Regions C and D are high-entropy zones caused by passage vortices at the rotor outlet near the upper and lower endwalls, where the loss due to the lower passage vortex is significantly reduced in the final design. Fig. 24 presents the efficiency distribution along the blade height for both designs. Compared to the reference design, the final design exhibits efficiency improvement over most of the blade span, particularly near the upper endwall and mid-span regions.

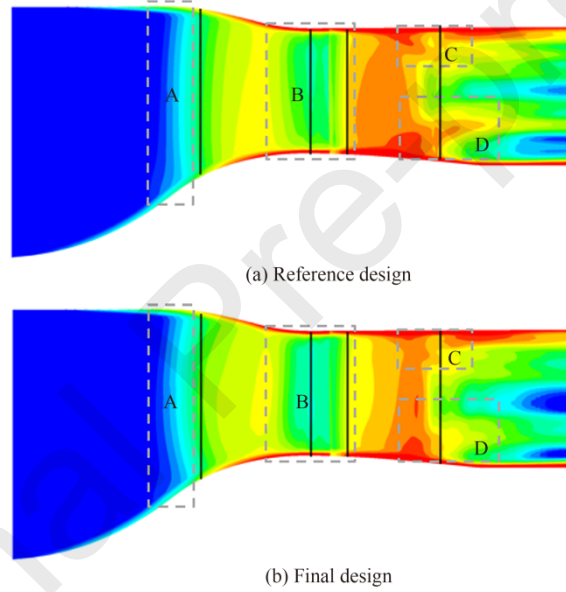


Fig. 23 Distribution of entropy on meridional plane

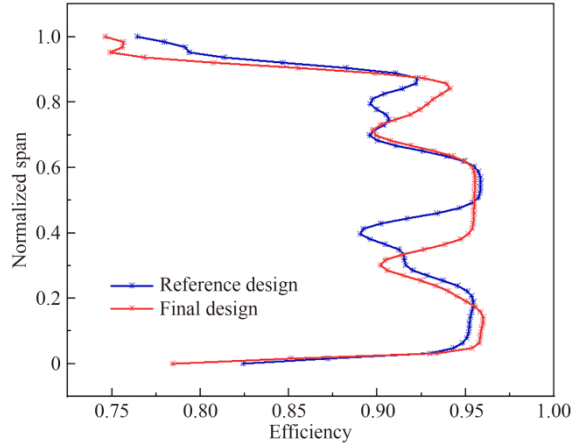


Fig. 24 Efficiency distribution curves along spanwise

5. Conclusions

This study established a SHAP-based data mining framework, upon which the overall performance of the design space, spanwise performance distribution, meridional physical field visualization, and knowledge mining of typical samples were conducted for the first-stage high-pressure turbine of the GE-E3 engine. Based on these analyses, design guidelines for the turbine stage were summarized and verified through aerodynamic validation. The main conclusions are as follows:

(1) For the turbine stage efficiency, the top three influential parameters are the blade effective outlet angle, the stator three-dimensional stacking parameters, and the spline control points near the suction-side leading edge. A reduced effective outlet angle at the rotor mid-span and an increased one at the stator mid-span both improve mid-span efficiency. Their combined effect enhances efficiency while limiting changes in mass flow rate.

(2) Among the three-dimensional stacking parameters of the stator, shifting the circumferential position at the blade tip toward the pressure side improves tip efficiency, while a similar shift at the mid-span enhances mid-span efficiency. These changes collectively form a positively leaned blade, which helps reduce radial pressure gradients within the vane passage and thereby improves turbine efficiency.

(3) Moving the spline control points near the suction-side leading edge at the mid-span of the stator and rotor in the direction of blade thinning is beneficial for reducing pressure gradients in the leading-edge acceleration region, decreasing irreversible flow losses, and improving efficiency across most of the span.

(4) The final design derived from the design guidelines achieves an efficiency of 91.09%, which is 0.65% higher than the reference design and exceeds the maximum value in the dataset, demonstrating the effectiveness and potential of SHAP-based analysis.

The SHAP-based data mining framework proposed in this study is highly generalizable and can be extended to knowledge discovery in multidisciplinary design spaces of various turbomachinery systems, offering useful guidance for similar design tasks.

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